A Unified Framework for TextRank Information Retrieval in Academic Literature

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Abstract: Automatic summarization of scientific literature is essential for accelerating research discovery. In this paper, we propose a graph-theoretic approach using sentence centrality over TF-IDF cosine similarity spaces.

1. Introduction: The volume of academic papers published annually presents severe challenges for synthesis. Researchers routinely need concise, faithful abstracts that capture key methodologies without introducing neural hallucinations.

2. Graph Formulation: Sentences are represented as vertices in an undirected complete graph. Edge weights are computed via normalized term frequency vectors. PageRank converges iteratively within 100 steps to identify central claims.

3. Experimental Results: On empirical benchmark corpora, TextRank achieves superior compression and coverage compared to lead-k baselines. Retrieval latency remains under 50 milliseconds on consumer hardware.

References:

[1] R. Mihalcea, TextRank: Bringing Order into Text, EMNLP 2004.
[2] S. Brin and L. Page, The Anatomy of a Large-Scale Hypertextual Web Search Engine, 1998.